CARDIO-ALPHA Renal Laboratory Substudy: Potassium and Creatinine

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Purpose

To characterize serum potassium and creatinine changes in a laboratory substudy of 214 CARDIO-ALPHA participants (108 Cardivex, 106 placebo).

Methods

Samples were collected at baseline, week 2, week 6 and week 12. Potassium was measured by indirect ion-selective electrode and creatinine by enzymatic assay; eGFR was calculated with the CKD-EPI 2021 equation.

Results

Mean serum potassium increased by 0.21 mmol/L with Cardivex versus 0.03 mmol/L with placebo at week 12. Mean serum creatinine increased by 6.2 umol/L (about 7%) at week 2 and stabilized thereafter. Three Cardivex patients (2.8%) had potassium above 5.5 mmol/L; all had baseline eGFR below 60 mL/min/1.73 m2.

Interpretation

The early creatinine rise is consistent with the expected hemodynamic effect of ACE inhibition and was not progressive.

Summary table

Parameter (week 12)	Cardivex (n=108)	Placebo (n=106)	Reference range
Potassium change, mmol/L	+0.21	+0.03	3.5-5.1
Creatinine change, umol/L	+6.2	+0.8	62-106 (M)
eGFR change, mL/min/1.73 m2	-3.9	-0.6	>=90